



Complex numbers - A1

Example ① $2x - 8 = 0$ $x = 4$
one solution

② $P(x) = x^2 - 25 = 0$ $x = 5, -5$
two solutions

③ $P(x) = x^2 + 1 = 0$ (No real solutions)
No solutions

Let i be a "imaginary" number
such that $i^2 = -1$

Def a complex number: $z = a + bi$
 $a, b \in \mathbb{R}$

$\mathbb{C}$ = all complex numbers

Ex

$$z_1 = 3 - 2i$$

$$z_2 = 4 + 5i$$

Def $z_1 = a_1 + b_1 i$, $z_2 = a_2 + b_2 i$

$$z_1 = z_2 \iff a_1 = a_2, b_1 = b_2$$

$$z = a + bi$$

$\text{Re}(z) = a$ real part of z

$\text{Im}(z) = b$ imaginary part of z

Ex

$$z = 7 - 8i \quad \text{Re}(z) = 7$$
$$\text{Im}(z) = -8$$

Operations

① addition

$$z_1 = a_1 + b_1 i$$

$$z_2 = a_2 + b_2 i$$

$$z_1 + z_2 = (a_1 + a_2) + (b_1 + b_2) i$$

Ex $(2-7i) + (3+8i) = 5 + i$

Note a real number is a complex number with imaginary part zero
 $\mathbb{R}$ is a subset $\mathbb{C}$

Complex Conjugate

$$z = a + bi \quad \bar{z} = a - bi$$

Ex $z = -4 - 7i \Rightarrow \bar{z} = -4 + 7i$

if $z = a$ (if $\text{Im}(z) = 0$)

$$\bar{\bar{z}} = a = z$$

$$(\bar{\bar{z}}) = z$$

multiplication

$$(a+b)(c+d) = ac + ad + bc + bd$$

$$(a_1 + bi)(a_2 + b_2i) = a_1a_2 + a_1b_2i + a_2b_1i + b_1b_2i^2$$

$$(but\ i^2 = -1)$$

$$= (a_1a_2 - b_1b_2) + (a_1b_2 + a_2b_1)i //$$

$$i^4 = (i^2)^2 = (-1)^2 = 1$$

$$i^3 = i^2 \cdot i = -1$$

Ex $(3-2i)(4+i) = 12 + 3i - 8i + 2 = 14 - 5i //$

Special case:

$$(3+4i)(3-4i) = 9 - \cancel{12i} + \cancel{12i} + 16 \\ = 25$$

$$(a+bi)(a-bi) = a^2 - abi + abi + b^2 \\ = a^2 + b^2$$

Def $Z = a + bi$

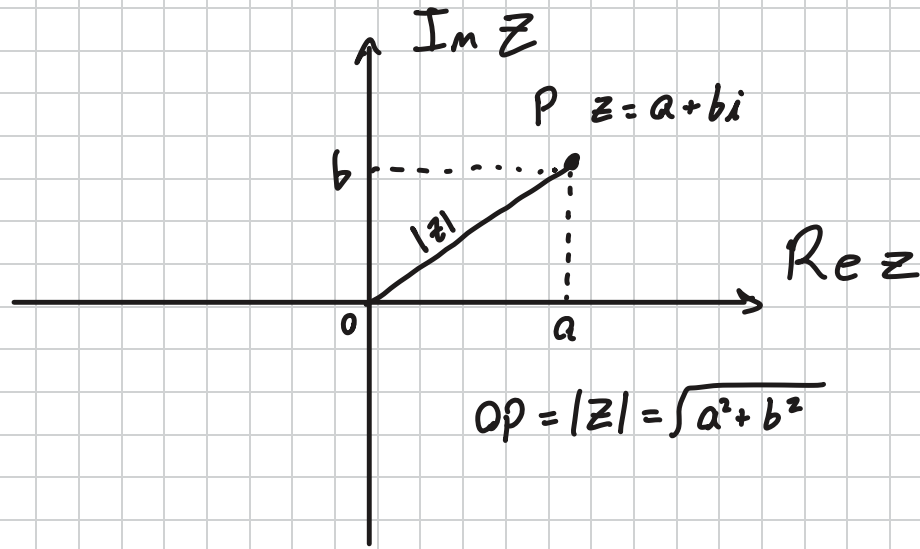
absolute
value of Z

$$|Z| = \sqrt{a^2 + b^2}$$

always a positive real number

$$Z \cdot \bar{Z} = |Z|^2, \quad |Z| = |\bar{Z}|$$

the complex plane



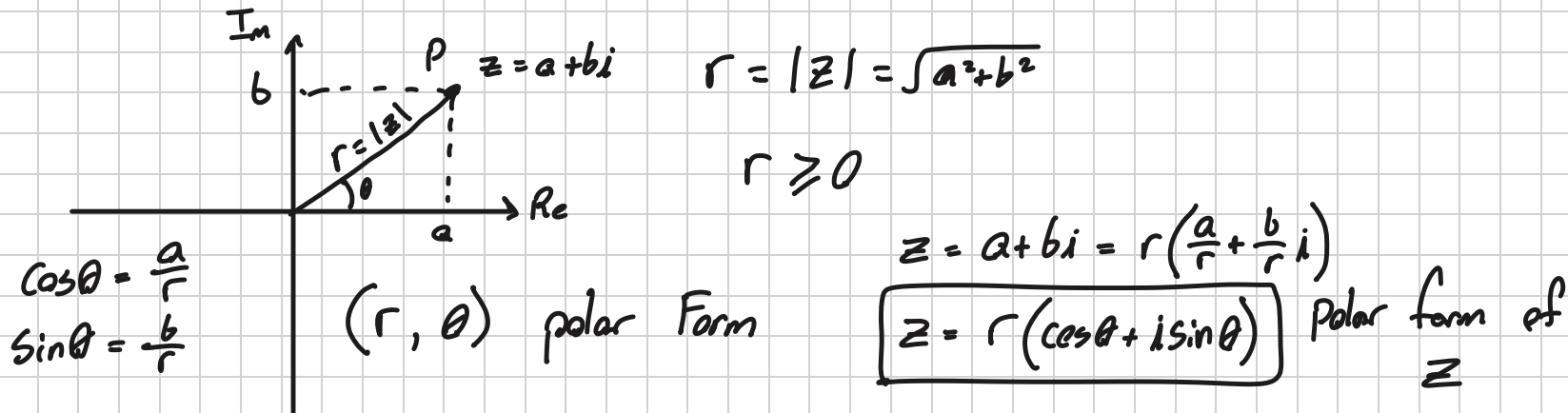
Division

$$\begin{aligned} \frac{3+4i}{2-i} &= \frac{3+4i}{2-i} \times \frac{2+i}{2+i} = \frac{(3+4i)(2+i)}{(2-i)(2+i)} = \frac{6+3i+8i-4}{5} = \frac{2+11i}{5} = \frac{2}{5} + \frac{11}{5}i \end{aligned}$$

→ multiply top and bottom by complex conjugate of the denominator.

$z = a + bi$ Cartesian form of a complex number

The polar Form



E_x

$$z = 1 + i$$

$$r = |1 + i| = \sqrt{2}, \quad \theta = \frac{\pi}{4}$$

$$z = \sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$